

5. Functional Requirements

5.1 User Management & Authentication

5.1.1 Internal Users

- SSO integration with Xebia's existing Identity Provider
- Automated user provisioning/deprovisioning via SCIM or HRIS API
- Profile management with role, department, location, and skills attributes
- Manager hierarchy support for approval workflows and reporting

5.1.2 External Users

- Self-registration with email verification for B2C learners
- Bulk enrolment / CSV upload for B2B client organisations
- Social login (Google, LinkedIn) for B2C
- Guest / preview access for course catalogue browsing (limited, unauthenticated)
- Multi-factor authentication (MFA) support
- Learner profile with photo, bio, learning history, and earned badges

5.2 Course & Content Management

- Support for SCORM 1.2, SCORM 2004, xAPI (Tin Can), and AICC content standards
- Native content builder supporting video, text, images, PDFs, PPT, and embedded media
- Video hosting with adaptive bitrate streaming (HLS), chapters, and transcripts
- Course versioning to maintain learner progress through updates
- Content tagging, categorisation, and metadata management for search and filtering
- Drag-and-drop course builder for instructors
- Multi-language content support (English minimum; Hindi, Arabic as Phase 2)
- Ability to embed third-party content (YouTube, LinkedIn Learning, Coursera, etc.)
- Document and resource library accessible by learners
- Course preview / demo content for external catalogue

5.3 Learning Paths & Programmes

- Ability to create structured learning paths (sequential or flexible) bundling multiple courses
- Prerequisite management—gate progression based on course completion or assessment scores
- Programme templates for onboarding, compliance, technical upskilling, and leadership
- Blended learning support: mix of self-paced, VILT (virtual instructor-led), and assessments
- Cohort-based learning for batch/intake-driven programmes (especially for University and B2B)
- Scheduled live sessions with calendar integration (Zoom, MS Teams)

5.4 Enrolment & Access Management

- Self-enrolment with manager approval workflow (configurable)
- Admin-assigned enrolment for mandatory training
- Waitlist management for capacity-limited programmes
- Enrolment rules based on role, department, or custom attributes
- Course access expiry with configurable duration (e.g., 30/60/90/365 days)
- Auto-enrolment triggers based on HRIS events (new joiner, role change)

5.5 Assessment & Evaluation Engine

- Multiple question types: MCQ, multi-select, true/false, fill in the blank, drag-and-drop, video response, coding challenge
- Question bank with randomisation and difficulty tagging
- Configurable pass/fail thresholds, attempt limits, and retake policies
- Timed assessments with auto-submission on timeout
- Anti-cheating features: question shuffling, browser lockdown mode (optional)
- Instructor-graded assignments with rubric builder and feedback capability
- Peer review / peer assessment workflows
- Integrated plagiarism check (API-based, e.g., Turnitin) for written submissions
- Post-course surveys and feedback forms (Kirkpatrick Level 1)

5.6 Certifications & Digital Badges

- Automated certificate generation on course or programme completion
- Customisable certificate templates with Xebia and client branding
- Certificate expiry and renewal reminders
- Open Badges 2.0 compliant digital badge issuance
- Badge sharing to LinkedIn and other social platforms
- Publicly verifiable certificate link / QR code
- Bulk certificate generation and download for admins

5.7 Analytics & Reporting

5.7.1 Learner Analytics

- Individual learner dashboard: progress, completions, scores, time spent, upcoming deadlines
- Course-level analytics: enrolment, completion rate, drop-off, assessment performance
- Skill gap analysis and competency mapping
- Learning streak and engagement scoring

5.7.2 Manager / L&D Admin Reports

- Team-level completion dashboards with drill-down by individual
- Compliance status tracking with overdue flagging and automated reminders
- Learning ROI and hours logged reports

- Cohort progress tracking for batch programmes
- Custom report builder with scheduled email delivery (CSV, PDF, Excel)
- Pre-built report templates for HR, compliance, and L&D use cases

5.7.3 Commercial / Revenue Analytics

- Revenue by course, client, learner segment, and region
- Subscription and licence utilisation reports
- Sales funnel and conversion analytics for course catalogue
- Refund, chargeback, and coupon redemption reports

5.8 Multi-Tenancy & White-Labelling

- Support for multi-tenant architecture with data isolation between tenants (client organisations)
- Each B2B client to have a branded sub-portal: custom logo, colour scheme, domain/subdomain
- Configurable email templates and notification branding per tenant
- Xebia internal portal fully branded with Xebia identity guidelines
- Ability to restrict catalogue visibility by tenant

5.9 Commerce & Monetisation

- Public course catalogue with pricing, course preview, ratings, and reviews
- Payment gateway integration: Razorpay, Stripe, PayPal (region-specific)
- Support for one-time purchase, subscription, and cohort/batch pricing models
- B2B invoice-based billing with purchase order (PO) and credit terms support
- Coupon codes, promotional pricing, and referral discounts
- Bundle/course package pricing
- Seat licence model for enterprise clients (e.g., 100 seats for Org X)
- GST / VAT compliance for India, UAE, and other applicable APMEA jurisdictions
- Automated invoicing and receipt generation
- Refund management workflow with configurable refund policies

5.10 University / Academic Module

- LTI 1.3 compliant integration with university LMS systems (Moodle, Canvas, Blackboard)
- Academic calendar-based cohort management with semester/term support
- Assignment submission, grading, and gradebook compatible with academic standards
- Academic integrity features: plagiarism detection, submission audit trail
- Faculty/instructor dashboard aligned to academic workflows
- Transcript generation in academic format

5.11 Communication & Notifications

- Automated email notifications: enrolment confirmation, reminders, completion, certificate issuance
- In-platform notifications and notification centre
- Configurable push notifications for mobile app
- Announcement and broadcast messaging from admin/instructor to learners
- Discussion forums and Q&A within courses
- Direct messaging between learners and instructors (within platform)
- Escalation alerts for compliance overdue to manager and HR

5.12 Mobile & Accessibility

- Fully responsive web design (mobile, tablet, desktop)
- Native mobile app (iOS and Android) or high-fidelity Progressive Web App (PWA)
- Offline content access for downloaded courses on mobile
- WCAG 2.1 Level AA accessibility compliance
- Screen reader compatibility, keyboard navigation, and captions on all video content

6. Technical Requirements

6.1 Architecture

- Cloud-native, multi-tier architecture
- Microservices-based design for modularity and independent scaling
- Multi-tenant SaaS architecture with logical data isolation
- Containerised deployment (Docker / Kubernetes)
- CDN for video and static asset delivery (CloudFront, Akamai, or equivalent)
- Headless / API-first design enabling future mobile and third-party integrations

6.2 Performance & Scalability

Parameter	Requirement
Concurrent Users	500+ concurrent users at launch; scalable to 5,000+
Page Load Time	< 3 seconds for all primary pages (P95)
Video Start Time	< 5 seconds buffering on standard broadband
API Response Time	< 500 ms for 95% of API calls
Uptime SLA	99.9% (excluding scheduled maintenance)
Maintenance Window	< 4 hours/month, with 72 hours advance notice
Horizontal Scaling	Auto-scaling based on load metrics
File Upload Limit	Up to 4 GB per file (video); 100 MB per document

6.3 Security & Compliance

- Data encryption at rest (AES-256) and in transit (TLS 1.2+)
- Role-based access control (RBAC) with principle of least privilege
- OWASP Top 10 compliance; regular penetration testing
- GDPR compliance for EU data subjects; compliance with applicable APMEA data protection laws (PDPA, etc.)
- Data residency options: ability to specify storage region (India, Singapore, UAE)
- Audit logging for all admin and security-relevant actions
- DDoS protection and WAF (Web Application Firewall)
- Regular vulnerability scanning and patching SLA < 72 hours for critical
- SOC 2 Type II certification preferred (or commitment to achieve within 12 months)

6.4 Integration Requirements

Integration	System / Standard	Priority	Notes
HR / People System	Workday, BambooHR, SAP HCM (via API/SCIM)	P1	User sync, auto-enrolment triggers

Integration	System / Standard	Priority	Notes
SSO / Identity	Azure AD, Okta, SAML 2.0, OAuth 2.0	P1	Internal employee auth
Video Conferencing	Zoom, Microsoft Teams	P1	VILT session scheduling & links
Payment Gateway	Razorpay, Stripe, PayPal	P1	B2C and B2B commerce
CRM	Salesforce, HubSpot (webhook/API)	P2	Lead capture, B2B enrolment
Content Providers	LinkedIn Learning, Coursera, Pluralsight	P2	Embedded or syndicated content
LTI Integration	LTI 1.3 standard	P1	University / academic partners
Email / Comms	SendGrid, AWS SES	P1	Transactional email delivery
Analytics	Google Analytics, Power BI, Tableau	P2	Advanced reporting integration
Digital Badges	Credly, Open Badges 2.0	P2	Badge issuance and verification

6.5 API & Developer Requirements

- RESTful API with OpenAPI 3.0 documentation
- Webhook support for key events (enrolment, completion, payment, badge issued)
- Rate limiting and API key management
- Sandbox / test environment with full API parity
- API versioning to ensure backward compatibility